Layer 13: Ratiometric Signal Processing and Enzyme Quantification

From Peak Areas to Enzyme Concentration with Propagated Uncertainty

Eton iGEM 2026, Dry Lab | Layer 13 of 17-Layer Biosensor Pipeline | `ratiometric.py`

Layer 12 delivers two peak areas per SWV scan: A_{MB} (from the MB redox label covalently attached to the peptide probe SGKIPRTLTA) and A_{Fc} (from the ferrocenemethanol internal reference). Layer 13 converts these two numbers into a single enzyme concentration estimate [E] with a full uncertainty budget.

The challenge this layer must solve is that the raw peak areas are corrupted by at least four distinct artefacts: variation in electrode area A_{true} between cartridge wells, gain drift in the transimpedance amplifier, variation in the SWV voltage amplitude ΔE_{sw} over time, and electrode fouling (non-specific adsorption of plasma proteins that blocks the electrode surface). The ratiometric scheme described here cancels three of these artefacts exactly and the fourth approximately, leaving a signal whose remaining uncertainty comes almost entirely from Poisson-like shot noise on the peak area fit, which can be propagated analytically.

I. The Raw Ratio R: Cancelling Electrode-to-Electrode Variation

Term-by-term derivation

From Layer 11 (Eq. 11.8), the measured MB peak area on electrode i is:

$$A_{\mathrm{MB}}^{(i)} = n_e^2 \frac{F^2}{2RT}\, f(\kappa_{\mathrm{SW}}^{(i)})\, A_{\mathrm{true}}^{(i)}\, \Gamma_{\mathrm{intact}}^{(i)}\, E_{\mathrm{sw}}^{(i)} \tag{I.1}$$

where $n_e = 2$, $f(\kappa_{\mathrm{SW}})$ is the dimensionless kinetic factor (between 0 and 1, from Layer 11, Section III), $A_{\mathrm{true}}^{(i)}$ is the actual gold area of electrode i, $\Gamma_{\mathrm{intact}}^{(i)}$ is the surviving intact probe surface coverage, and $E_{\mathrm{sw}}^{(i)}$ is the SWV amplitude delivered by the potentiostat at the time of the scan.

The ferrocenemethanol reference peak area, from the same electrode and the same scan, obeys the same equation with $n_e = 1$, its own formal potential, its own kinetic factor, its own tether length (set by design to give $f(\kappa_{\mathrm{SW}}^{\mathrm{Fc}}) \approx 1$ at the chosen f_{SWV}), and its own surface coverage $\Gamma_{\mathrm{Fc}}^{(i)}$:

$$A_{\mathrm{Fc}}^{(i)} = 1 \cdot \frac{F^2}{2RT}\, f(\kappa_{\mathrm{SW}}^{\mathrm{Fc},(i)})\, A_{\mathrm{true}}^{(i)}\, \Gamma_{\mathrm{Fc}}^{(i)}\, E_{\mathrm{sw}}^{(i)} \tag{I.2}$$

Notice that $A_{\mathrm{true}}^{(i)}$ and $E_{\mathrm{sw}}^{(i)}$ appear identically in both Eq. (I.1) and Eq. (I.2). The ratio:

$$\boxed{R^{(i)} = \frac{A_{\mathrm{MB}}^{(i)}}{A_{\mathrm{Fc}}^{(i)}} = n_e^2\, \frac{f(\kappa_{\mathrm{SW}}^{(i)})}{f(\kappa_{\mathrm{SW}}^{\mathrm{Fc},(i)})}\, \frac{\Gamma_{\mathrm{intact}}^{(i)}}{\Gamma_{\mathrm{Fc}}^{(i)}}} \tag{I.3}$$

The electrode area $A_{\mathrm{true}}^{(i)}$ and the SWV amplitude $E_{\mathrm{sw}}^{(i)}$ cancel exactly. Physical constants F, R, T also cancel, removing the need for temperature calibration in this ratio (though Layer 10 still needs T for the $E_{\mathrm{MB}}^{\circ\prime}(\mathrm{pH})$ prediction).

What R still depends on

The remaining factors in Eq. (I.3) are: (1) The kinetic ratio $f(\kappa_{\mathrm{SW}})/f(\kappa_{\mathrm{SW}}^{\mathrm{Fc}})$. Because the Fc tether is chosen to put Fc in the fast-kinetics limit ($\kappa_{\mathrm{SW}}^{\mathrm{Fc}} \gg 1$ so $f \approx 1$), and because the MB kinetic factor $f(\kappa_{\mathrm{SW}})$ is a function of $k_{\mathrm{ET}}^0 \times t_{\mathrm{pulse}}$ that is the same for all electrodes of the same design (since k_{ET}^0 depends on tether length, not on A_{true}), this ratio is a device-level constant that cancels when we subsequently take $R(T)/R(0)$ (Section II).

(2) The coverage ratio $\Gamma_{\mathrm{intact}}/\Gamma_{\mathrm{Fc}}$. The Fc SAM and the aptamer-MB SAM are co-deposited, so their surface densities are *assumed* to be correlated: electrodes with more Fc also have more MB. If the Fc coverage is stable (Fc molecules do not desorb during the assay, as confirmed by Layer 12 diagnostics), then Γ_{Fc} is a constant for a given electrode and the ratio $\Gamma_{\mathrm{intact}}/\Gamma_{\mathrm{Fc}}$ evolves purely due to changes in Γ_{intact} (i.e., cleavage or folding of the aptamer probe).

Status of this assumption on the current device (FLAG). White and Plaxco [1] validated this cancellation on a different aptamer-beacon system, not on our co-immobilised probe-peptide/C6FC SAM. It has not been independently confirmed for this device, and preliminary internal data suggests it may *not* show the clean cancellation this section assumes [5]. The co-deposition correlation above should therefore be treated as an unverified working

assumption, not an established result, until tested directly (e.g. by deliberately varying total probe coverage electrode-to-electrode and checking whether R still tracks $\Gamma_{\text{intact}}/\Gamma_{\text{Fc}}$ as predicted).

White and Plaxco [1] confirmed experimentally that taking this ratio eliminates electrode-to-electrode amplitude variation by over 10-fold in a chip with deliberate variation in electrode size, validating the A_{true} cancellation on their system.

Uncertainty on R

The uncertainty on R from the peak fitting uncertainties is propagated using the standard formula for the ratio of two independent random variables:

$$\left(\frac{\sigma_R}{R}\right)^2 = \left(\frac{\sigma_{A_{\text{MB}}}}{A_{\text{MB}}}\right)^2 + \left(\frac{\sigma_{A_{\text{Fc}}}}{A_{\text{Fc}}}\right)^2 \tag{I.4}$$

This assumes that the MB and Fc areas are independent; they are indeed fitted from different non-overlapping regions of the voltammogram (Section 12.V), so their fit errors are uncorrelated. The relative uncertainty on R is therefore quadrature-added from the relative uncertainties on each area.

Numerically, with $A_{\text{MB}} \approx 0.3$ nA V and $\sigma_{A_{\text{MB}}} \approx 3\%$ of A_{MB} (Layer 12, Eq. 12.12), and with similar relative precision on A_{Fc}: $\sigma_R/R \approx 4.2\%$ per scan.

II. The Normalised Signal ρ: Cancelling Probe Density Variation

Motivation

Even after forming R (Eq. I.3), the coverage ratio $\Gamma_{\text{intact}}/\Gamma_{\text{Fc}}$ varies between electrodes (different deposition conditions, different self-assembly kinetics) by up to $\pm 30\%$ (Lubin and Plaxco [2]). This means $R(0)$ (the ratio measured before enzyme exposure) is electrode-specific. To cancel this, we normalise each electrode's $R(T)$ at time T by its own baseline $R(0)$:

$$\boxed{\rho^{(i)}(T) = \frac{R^{(i)}(T)}{R^{(i)}(0)} = \frac{\Gamma^{(i)}_{\text{intact}}(T)}{\Gamma^{(i)}_{\text{intact}}(0)}} \tag{II.1}$$

The kinetic ratio and Fc coverage cancel in Eq. (II.1) because they are constant for a given electrode: $f(\kappa_{\text{SW}})$ does not change during the assay (the scan parameters are fixed), and Γ_{Fc} is stable over the 30-minute assay time (Layer 12 diagnostics confirm $< 2\%$ Fc amplitude drift per scan).

The physical meaning of ρ is exactly the fraction of aptamer probes that remain intact at time T: $\rho = 1$ means no cleavage, $\rho = 0$ means complete cleavage. For the biosensor, we expect $\rho \in [0.1, 1]$ over the clinically relevant enzyme concentration range.

Concrete fouling example

Consider an electrode with 20% non-specific protein adsorption (fouling) that covers a fraction of the gold surface. Fouling reduces both A_{MB} and A_{Fc} by the same fractional amount f_{foul}:

$$A_{\text{MB}}(T) = A^{\text{clean}}_{\text{MB}}(T) \times (1 - f_{\text{foul}}) \tag{II.2}$$

$$A_{\text{Fc}}(T) = A^{\text{clean}}_{\text{Fc}}(T) \times (1 - f_{\text{foul}}) \tag{II.3}$$

When we form $R = A_{\text{MB}}/A_{\text{Fc}}$, the fouling factor $(1 - f_{\text{foul}})$ cancels because it appears in both numerator and denominator. Similarly, the baseline measurement $R(0)$ also suffered the same fouling (adsorption is a fast process that saturates before the first scan). Therefore $\rho = R(T)/R(0)$ also cancels the fouling factor: the ratio of the ratios is the ratio of the cleavage fractions with no fouling term surviving.

This cancellation is *exact* only if the fouling is the same on the Fc and MB regions of the electrode, which is true if the SAM is uniform. A non-uniform fouling pattern (e.g., a protein adsorbing preferentially to the aptamer-MB SAM but not to the Fc SAM because of electrostatic differences) would cause only partial cancellation. Section VII addresses this limitation.

Uncertainty on ρ

$$\left(\frac{\sigma_\rho}{\rho}\right)^2 = \left(\frac{\sigma_{R(T)}}{R(T)}\right)^2 + \left(\frac{\sigma_{R(0)}}{R(0)}\right)^2 \tag{II.4}$$

The baseline $R(0)$ is the average of three scans taken before enzyme addition; averaging $n_0 = 3$ scans reduces $\sigma_{R(0)}$ by $1/\sqrt{3}$. With $\sigma_R/R \approx 4.2\%$ per scan and three baseline scans:

$$\frac{\sigma_{R(0)}}{R(0)} = \frac{4.2\%}{\sqrt{3}} = 2.4\% \tag{II.5}$$

Combining with a single on-assay scan:

$$\frac{\sigma_\rho}{\rho} = \sqrt{(4.2\%)^2 + (2.4\%)^2} = 4.8\% \tag{II.6}$$

This is the irreducible per-electrode fractional uncertainty on ρ. Taking the average over N_{active} active electrodes on a chip reduces it further by $1/\sqrt{N_{\text{active}}}$.

III. The Double Ratio F_k: Cancelling Cleavage Independent of Enzyme

Background cleavage and the control electrode

Even without target enzyme, the aptamer probe is not perfectly stable. Background cleavage can occur from (a) dissolved nucleases in the sample matrix (plasma contains substantial nuclease activity), (b) hydrolysis of the DNA backbone at the assay temperature, and (c) reactive oxygen species in the wound fluid.

This background cleavage reduces ρ on the active electrode by an amount $\Delta\rho_{\text{bg}}$ that is unrelated to the target enzyme. Unless accounted for, it would overestimate F_k and therefore overestimate [E].

The solution is a *control electrode*: an electrode bearing the D-amino acid enantiomer of the peptide substrate (D-SGKIPRTLTA) in an otherwise identical SAM. MMP-9, like all natural metalloproteinases, is stereospecific for L-amino acid substrates: the active-site geometry that positions the scissile bond over the catalytic zinc ion requires the L-configuration at each α-carbon. The D-enantiomer substrate cannot adopt this geometry and is therefore not cleaved by MMP-9 (or any natural L-specific protease in wound fluid) regardless of enzyme concentration. Non-stereospecific background processes (non-specific nucleases acting on the thiol linker, gold–thiol bond hydrolysis, and reactive oxygen species) act on both the L-peptide active electrode and the D-peptide control equally. The control electrode therefore reports $\rho_{\text{control}}(T)$ as the surviving fraction under background conditions alone.

Derivation of F_k

The enzyme-specific fractional cleavage is:

$$\boxed{F_k = 1 - \frac{\rho_{\text{active}}(T)}{\rho_{\text{control}}(T)}} \tag{III.1}$$

To understand what cancels in Eq. (III.1), write the surviving fractions explicitly. Let p_{bg} be the fraction of probes cleaved by background mechanisms, and p_{E} be the additional fraction cleaved by the target enzyme. Then:

$$\rho_{\text{active}} = (1 - p_{\text{bg}})(1 - p_{\text{E}}) \tag{III.2}$$

$$\rho_{\text{control}} = (1 - p_{\text{bg}}) \tag{III.3}$$

Substituting into Eq. (III.1):

$$F_k = 1 - \frac{(1 - p_{\text{bg}})(1 - p_{\text{E}})}{(1 - p_{\text{bg}})} = 1 - (1 - p_{\text{E}}) = p_{\text{E}} \tag{III.4}$$

The factor $(1 - p_{\text{bg}})$ cancels exactly, leaving $F_k = p_{\text{E}}$: the fraction of active probes cleaved by the target enzyme alone, regardless of what background cleavage occurred. This is the quantity that directly constrains [E] through the kinetic model.

Lubin and Plaxco [2] validate this background cancellation experimentally by showing that control electrodes in the same plasma sample track active electrodes when no target is added, confirming that the ratio $\rho_{\text{active}}/\rho_{\text{control}}$ is stable at ≈ 1 in the absence of enzyme.

What does not cancel in F_k

Two sources of artefact are not cancelled by Eq. (III.1):

(1) *Differential fouling.* If the active and control SAMs foul differently (Section II), the Fc amplitude on the two electrodes drifts at different rates. The ratio $\rho_{\text{active}}/\rho_{\text{control}}$ then drifts even without enzyme. This is controlled by monitoring the Fc peak amplitude on both electrodes; a difference $> 10\%$ flags differential fouling and the control correction is invalidated.

(2) *Non-stereospecific metalloprotease activity.* Although natural MMPs are L-stereospecific, some bacterial proteases and heat-denatured MMPs may exhibit reduced stereospecificity. Any MMP-9 activity on the D-enantiomer would cause ρ_{control} to decrease with enzyme concentration, causing Eq. (III.4) to underestimate p_{E}. Validation requires adding saturating MMP-9 to a chip bearing only D-enantiomer control electrodes and confirming ρ_{control} drops by less than 5%.

Uncertainty on F_k
From Eq. (III.1):

$$\sigma_{F_k}^2 = \left(\frac{1}{\rho_{\text{control}}}\right)^2 \sigma_{\rho_{\text{active}}}^2 + \left(\frac{\rho_{\text{active}}}{\rho_{\text{control}}^2}\right)^2 \sigma_{\rho_{\text{control}}}^2 \tag{III.5}$$

If active and control electrodes have equal per-electrode uncertainty $\sigma_\rho/\rho \approx 4.8\%$ (Eq. II.6) and if $\rho_{\text{active}} \approx \rho_{\text{control}} \approx \rho$ (early time point, small cleavage), then:

$$\sigma_{F_k} \approx (1 - F_k)\,\sqrt{2}\,\frac{\sigma_\rho}{\rho} = (1 - F_k) \times 6.8\% \tag{III.6}$$

This expression contains the factor $(1 - F_k)$, which decreases toward zero as $F_k \to 1$ (complete cleavage). It might seem that the uncertainty also vanishes, but this is misleading: the fractional uncertainty $\sigma_{F_k}/(1 - F_k)$ is actually $\sqrt{2} \times \sigma_\rho/\rho = 6.8\%$, constant regardless of F_k. The practical limit of detection is set by whether F_k exceeds $3\sigma_{F_k}$ at zero enzyme concentration, i.e., whether $3 \times 6.8\% \approx 20\%$ cleavage can be reliably distinguished from baseline noise.

IV. Inversion to Enzyme Concentration

Connecting F_k to the kinetic model
From Layer 9 (Section IV), the integrated MM equation gives the fraction of probes cleaved $F_k = p_{\text{E}}$ after incubation time T:

$$k_{\text{cat}}[\text{E}]\,T = K_{m,\text{surf}}^{\Gamma} \ln\!\left(\frac{1}{1 - F_k}\right) + \Gamma_{\text{intact}}(0)\,F_k \tag{IV.1}$$

where k_{cat} is the catalytic rate constant, [E] is the enzyme molar concentration, T is the incubation time, $K_{m,\text{surf}}^{\Gamma}$ is the surface Michaelis constant, and $\Gamma_{\text{intact}}(0)$ is the initial probe density.

Two limiting regimes and the apparent rate
In the unsaturated regime ($F_k \ll 1$, i.e., early time or low enzyme), Eq. (IV.1) simplifies. Using $\ln(1/(1-u)) \approx u + u^2/2$ for small u:

$$k_{\text{cat}}[\text{E}]\,T \approx \left(K_{m,\text{surf}}^{\Gamma} + \Gamma_{\text{intact}}(0)\right) F_k \quad \Rightarrow \quad F_k \approx \frac{k_{\text{cat}}[\text{E}]\,T}{K_{m,\text{surf}}^{\Gamma} + \Gamma_{\text{intact}}(0)} \tag{IV.2}$$

In this regime, F_k is linear in $[\text{E}] \times T$, making calibration straightforward: the slope is $k_{\text{cat}}/(K_{m,\text{surf}}^{\Gamma} + \Gamma_{\text{intact}}(0))$.

In the saturated regime ($\Gamma_{\text{intact}} \ll K_{m,\text{surf}}^{\Gamma}$, i.e., low probe density), the linear term dominates:

$$k_{\text{cat}}[\text{E}]\,T \approx K_{m,\text{surf}}^{\Gamma} F_k \quad \Rightarrow \quad F_k \approx \frac{k_{\text{cat}}[\text{E}]\,T}{K_{m,\text{surf}}^{\Gamma}} \tag{IV.3}$$

In both limits, we can define an *apparent rate* that depends only on observables:

$$\boxed{\kappa_k = -\frac{\ln(1 - F_k)}{T}} \tag{IV.4}$$

κ_k has units of s^{-1} (or min^{-1} for T in minutes). It is the rate of the monoexponential process that would produce the same F_k in time T. In the unsaturated regime where $F_k \ll 1$, $\ln(1/(1 - F_k)) \approx F_k$, so $\kappa_k \approx F_k/T = k_{\text{cat}}[\text{E}]/(K_{m,\text{surf}}^{\Gamma} + \Gamma_{\text{intact}}(0))$, which is linear in [E]. More generally, κ_k is a monotone increasing function of F_k and therefore of [E]: the calibration curve [E] vs κ_k is monotone and invertible.

Equation (IV.4) was chosen as the primary reported observable rather than F_k itself for two reasons. First, if the incubation time T varies between runs (e.g., because the clinician adds sample at a different time), κ_k normalises by T so the calibration curve is time-independent. Second, Lubin and Plaxco [2] and Baker 2006 [3] both show that the log-transformed observable is more linearly related to [E] across a wide dynamic range than F_k itself.

Dynamic range
The minimum detectable F_k is $F_k^{\text{min}} = 3\sigma_{F_k}(0) \approx 20\%$ (Section III). The maximum usable F_k is approximately 0.95 (near-complete cleavage, beyond which the MB signal is indistinguishable from noise). The corresponding apparent rate range is:

$$\kappa_k^{\text{min}} = \frac{-\ln(1 - 0.20)}{T} = \frac{0.223}{T}, \qquad \kappa_k^{\text{max}} = \frac{-\ln(1 - 0.95)}{T} = \frac{3.00}{T} \tag{IV.5}$$

giving a dynamic range of $3.00/0.223 = 13.5\times$ in apparent rate, or approximately 1.1 decades in κ_k. In the linear (unsaturated) regime where $\kappa_k \approx k_{\text{cat}}[\text{E}]/(K_{m,\text{surf}}^{\Gamma} + \Gamma_{\text{intact}}(0))$, κ_k is proportional to [E], so 1.1 decades in κ_k

corresponds to 1.1 decades in [E]. The clinically relevant MMP-9 range is 6–30 nM (chronic vs healing wound, Layer 0), spanning 0.7 decades. This fits comfortably within the 1.1-decade window, confirming that the biosensor can discriminate healing from chronic wounds across the full physiological range with the stated noise floor.

Inversion algorithm

Given a measured F_k and its uncertainty σ_{F_k}, the enzyme concentration is extracted by solving Eq. (IV.1) for [E]:

$$[\mathrm{E}] = \frac{1}{k_{\mathrm{cat}}\, T}\left[K^{\Gamma}_{m,\mathrm{surf}} \ln\!\left(\frac{1}{1-F_k}\right) + \Gamma_{\mathrm{intact}}(0)\, F_k\right] \tag{IV.6}$$

This is a closed-form expression: given the kinetic parameters k_{cat}, $K^{\Gamma}_{m,\mathrm{surf}}$, $\Gamma_{\mathrm{intact}}(0)$, and the measured F_k, no numerical solver is needed.

The uncertainty on [E] is:

$$\sigma_{[\mathrm{E}]} = \frac{1}{k_{\mathrm{cat}}\, T}\left(\frac{K^{\Gamma}_{m,\mathrm{surf}}}{1-F_k} + \Gamma_{\mathrm{intact}}(0)\right)\sigma_{F_k} \tag{IV.7}$$

The prefactor $K^{\Gamma}_{m,\mathrm{surf}}/(1-F_k) + \Gamma_{\mathrm{intact}}(0)$ diverges as $F_k \to 1$, reflecting the fact that near complete cleavage, a small absolute change in F_k corresponds to a large relative change in the number of remaining probes and therefore to a large change in the inferred concentration. In practice, readings with $F_k > 0.90$ are flagged as "above range": [E] is reported as "$> [\mathrm{E}]_{\max}$" rather than a precise value with a diverging uncertainty.

V. Error Propagation Chain

The full error propagation from raw peak areas to enzyme concentration is:

$$\underbrace{\sigma_{A_{\mathrm{MB}}},\, \sigma_{A_{\mathrm{Fc}}}}_{\text{Layer 12}} \xrightarrow{\text{Eq. I.4}} \sigma_R \xrightarrow{\text{Eq. II.4}} \sigma_\rho \xrightarrow{\text{Eq. III.6}} \sigma_{F_k} \xrightarrow{\text{Eq. IV.7}} \sigma_{[\mathrm{E}]} \tag{V.1}$$

Each arrow applies a specific formula derived above. The chain can be run forward (from measured areas to final uncertainty) or in reverse (from desired $\sigma_{[\mathrm{E}]}$ to required per-scan precision $\sigma_{A_{\mathrm{MB}}}$). The reverse direction is used during instrument design to set the required transimpedance amplifier noise floor.

Numerically, starting from $\sigma_{A_{\mathrm{MB}}}/A_{\mathrm{MB}} = 3\%$ per scan, the chain gives (at $F_k = 0.5$, $T = 30$ min, $N_{\mathrm{active}} = 4$ electrodes averaged):

$$\begin{aligned}
\sigma_R/R &= 4.2\% \ \text{(single scan)} \\
\sigma_\rho/\rho &= 4.8\% \ \text{(single electrode)} \\
&= 2.4\% \ \text{(4 electrodes)} \\
\sigma_{F_k} &= (1-0.5)\times\sqrt{2}\times 2.4\% = 1.7\% \\
\sigma_{[\mathrm{E}]} &\approx \frac{K^{\Gamma}_{m,\mathrm{surf}} + \Gamma_{\mathrm{intact}}(0)}{k_{\mathrm{cat}} T(1-0.5)} \times 1.7\% \ \text{(using Eq. IV.7)}
\end{aligned}$$

A 1.7% absolute uncertainty on $F_k = 0.5$ corresponds to approximately a 3.4% relative uncertainty on F_k at the midpoint, translating to approximately 3–5% relative uncertainty on [E] (varying with the exact slope of the calibration curve at that concentration).

VI. Electrode Layout and Averaging

The sensor chip contains a fixed number of electrodes per patient cartridge. The ratiometric pipeline aggregates readings from multiple electrodes of the same type using a weighted mean:

$$\bar{\kappa_k} = \frac{\sum_i w_i\, \kappa_k^{(i)}}{\sum_i w_i}, \qquad w_i = \frac{1}{\sigma^2_{\kappa_k^{(i)}}} \tag{VI.1}$$

where the weights are the inverse variances of the per-electrode apparent rates. Electrodes flagged by Layer 12 quality diagnostics are assigned $w_i = 0$. The uncertainty on $\bar{\kappa_k}$ is:

$$\sigma^2_{\bar{\kappa_k}} = \frac{1}{\sum_i w_i} \tag{VI.2}$$

The active-to-control electrode pairing is done before averaging: each active electrode i is paired with the spatially nearest control electrode j on the chip (to minimise differential fouling, which can have spatial gradients across the chip). The double ratio $F_k^{(i)}$ uses $\rho^{(j)}_{\mathrm{control}}$ from its paired control, not the mean of all controls.

Minimum electrode count and redundancy

The minimum number of active electrodes needed to resolve [E] at the LOD ($F_k = 0.20 \pm 3\sigma_{F_k}$) at a 95% confidence level is determined by requiring:

$$3\,\frac{\sigma_{F_k}}{\sqrt{N_{\text{active}}}} < F_k^{\min} \quad \Rightarrow \quad N_{\text{active}} > \frac{9\,\sigma_{F_k}^2}{(0.20)^2} = \frac{9 \times (0.068)^2}{0.04} \approx 1.0 \tag{VI.3}$$

A single active electrode is sufficient at the LOD (with $\sigma_{F_k} = 6.8\%$ from Eq. III.6). In practice, $N_{\text{active}} \geq 3$ is required for redundancy (in case one electrode fails the Layer 12 quality diagnostics) and to achieve $\sigma_{[\text{E}]}/[\text{E}] < 5\%$ at mid-range.

VII. Limitations

1. **The kinetic parameters k_{cat}, $K_{m,\text{surf}}^{\Gamma}$, and $\Gamma_{\text{intact}}(0)$ are not measured per-cartridge.** Equation (IV.6) requires these values as inputs. They are taken from Layer 9 (where k_{cat} and $K_{m,\text{surf}}^{\Gamma}$ come from bulk solution kinetics) and from Layer 11 (where $\Gamma_{\text{intact}}(0)$ is estimated from the MB peak area before cleavage using the Layer 11 equation). Any batch-to-batch variation in aptamer synthesis yield, thiol coupling efficiency, or gold deposition that shifts $\Gamma_{\text{intact}}(0)$ outside the range $[1 \times 10^{-12},\, 5 \times 10^{-12}]$ mol cm^{-2} will bias the inversion. A per-cartridge measurement of $\Gamma_{\text{intact}}(0)$ from $R(0)$ using the Layer 11 calibration is required for absolute accuracy.

2. **The double-ratio cancellation of fouling is exact only if fouling is uniform across the chip surface.** Baker 2006 [3] and Lubin and Plaxco [2] note that plasma protein adsorption varies with SAM composition. If the Fc-thiol region and the aptamer-thiol region adsorb proteins at different rates, the Fc amplitude on the active and control electrodes will drift differentially, violating the approximation in Section II. The Fc amplitude monitoring diagnostic (Layer 12) detects but does not correct this failure mode.

3. **The inversion formula Eq.** (IV.6) **assumes the integrated MM model, which uses surface $K_{m,\text{surf}}^{\Gamma}$ estimated from bulk K_m and an assumed effective diffusion layer depth.** Layer 9 Section IV establishes that $K_{m,\text{surf}}^{\Gamma}$ spans two orders of magnitude depending on the assumed δ_{eff}. If $K_{m,\text{surf}}^{\Gamma}$ is uncertain by a factor of 10, the inversion maps F_k to the wrong [E] by approximately the same factor in the $K_{m,\text{surf}}^{\Gamma}$-dominated regime ($F_k \ll 1$). Calibration with known enzyme concentration samples eliminates this systematic error empirically; the model then correctly predicts the *shape* of the calibration curve even if the absolute scale is set by the calibrant.

4. **The uncertainty formula Eq.** (III.6) **assumes Gaussian-distributed fit residuals.** If the baseline is non-polynomial (blocked by the Layer 12 quality diagnostic) or if the peak is non-Gaussian (double-MB population), the residuals are non-Gaussian and the propagated σ_{F_k} underestimates the true uncertainty.

5. **The D-enantiomer control must be verified stereospecifically resistant to MMP-9.** The design assumes MMP-9 cannot cleave D-SGKIPRTLTA. This assumption holds for natural L-specific metalloproteinases but must be confirmed experimentally: add saturating MMP-9 to a chip bearing only D-enantiomer electrodes and confirm ρ_{control} does not drop by more than 5%. Any residual activity on the D-enantiomer causes F_k from Eq. (III.4) to underestimate p_{E}.

References

All papers in `electrochemical_device/papers/layer_13/`.

[1] White RJ, Plaxco KW (2010). Exploiting binding-induced changes in probe flexibility for the optimization of electrochemical biosensors. *Anal Chem* 82:73–76. DOI: 10.1021/ac9028595. **white2010.pdf** – ratiometric signal construction, cancellation of electrode area variation, and SWV frequency optimisation confirmed.

[2] Lubin AA, Plaxco KW (2010). Folding-based electrochemical biosensors: the case for responsive nucleic acid architectures. *Acc Chem Res* 43:496–505. DOI: 10.1021/ar900165x. **lubin_plaxco2010.pdf** – structure-switching E-AB biosensor architecture, background cleavage correction by control electrode, fouling cancellation via double normalisation, and dynamic range characteristics confirmed.

[3] Baker BR, Lai RY, Wood MS, Doctor EH, Heeger AJ, Plaxco KW (2006). An electronic, aptamer-based small-molecule sensor for the rapid, label-free detection of cocaine in adulterated samples and biological fluids. *J Am Chem Soc* 128:3138–3139. DOI: 10.1021/ja056957p. **baker2006.pdf** – on-chip reference electrode stability, SWV parameter validation, and dynamic range in biological fluid confirmed.

[4] Lubin AA, Lai RY, Baker BR, Heeger AJ, Plaxco KW (2010). Electrochemical, structure-switching biosensors: quantitative analysis of the usable dynamic range. *Anal Chem* 78:5671–5677. DOI: 10.1021/ac0601417. **lubin2010.pdf** – per-electrode $R(0)$ baseline normalisation, electrode-to-electrode $R(0)$ variation, and Fc surface coverage stability during assay confirmed.

[5] Dry lab team, internal electrode characterisation notes (September 2026): preliminary evidence that the MB/Fc ratio may not cancel electrode-to-electrode coverage drift as cleanly as assumed on this device.